SMART MENU SYSTEM ARCHITECTURE

Service Window QR Codes, Table Identification & Fraud Prevention Strategy

Project: smart-menu-main | Date: September 2026

1. Executive Summary

This architectural report provides a comprehensive evaluation of introducing a Service Window (Counter / Takeaway / Kiosk) QR Code alongside traditional Table QR Codes in the Smart Menu system. It outlines technical workflows, table identification strategies, fraud prevention mechanisms, and code modification guidelines tailored for the smart-menu-main codebase.

2. Service Window QR Code Overview

A Service Window QR code allows customers entering the venue, waiting in line, or standing at an order counter to immediately scan and view the menu without first sitting down at a specific table.

- Purpose: Accelerates menu browsing, enables takeaway/counter ordering, and reduces waiter response time.
- Token Structure: Encodes a JWT payload with table_id = 'COUNTER' or 'SERVICE_WINDOW' (e.g. {"menu_id": "m1", "table_id": "COUNTER"}).
- Verification Flow: Scanned via ScanVerifier.jsx and validated against the backend /api/verify-scan endpoint.

3. Table Identification Mechanisms

When an order originates from a Service Window QR code, the system must determine where the food should be served. Three distinct models can be supported:

Model A: Counter Pickup / Order Callout (No Table Assigned)
The order is tagged with table_id = 'COUNTER'. When prepared, the kitchen updates the status to 'ready' on the admin portal. Customers collect their food from the window when their Order Number is displayed or called out.

Model B: Manual Table Input at Checkout
The customer scans the Service Window QR, builds their cart, and selects 'Dine-In' at checkout. A modal prompts them to enter their Table Number (e.g., Table 1â 20). The order payload posts customer_selected_table_id to /api/order.

Model C: Two-Step Dynamic Table Pairing (Recommended)
The customer scans the Service Window QR while standing in line to build a cart. Once seated, they tap 'Pair Table' inside the web app and scan the physical Table QR code. The session token upgrades from 'COUNTER' to 'TABLE_5'.

4. Security & Anti-Abuse Strategy

Allowing customers to order without being pre-locked to a table introduces potential fraud risks (e.g., placing fake orders to random tables). Below is the recommended multi-layered security strategy: